

# Inheritance

Polymorphism, Subtyping, Reuse

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Lecture #8 out of 8  
80 minutes

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Polymorphism

Implementation Inheritance

Chapter #1:

# Polymorphism

[ LSP SOLID Subtyping Generics Overloading ]

## Liskov Substitution Principle (LSP)

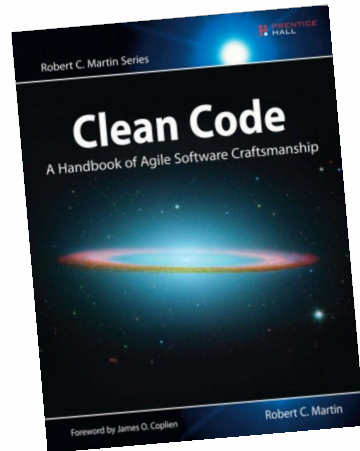


“If for each object  $o_1$  of type  $S$  there is an object  $o_2$  of type  $T$  such that for all programs  $P$  defined in terms of  $T$ , the behavior of  $P$  is unchanged when  $o_1$  is substituted for  $o_2$ , then  $S$  is a subtype of  $T$ .”

— Barbara Liskov. *Keynote Address — Data Abstraction and Hierarchy*. Proceedings of the Object-Oriented Programming Systems, Languages and Applications (Addendum), 1987

[ LSP SOLID Subtyping Generics Overloading ]

## SOLID (the “L” part)



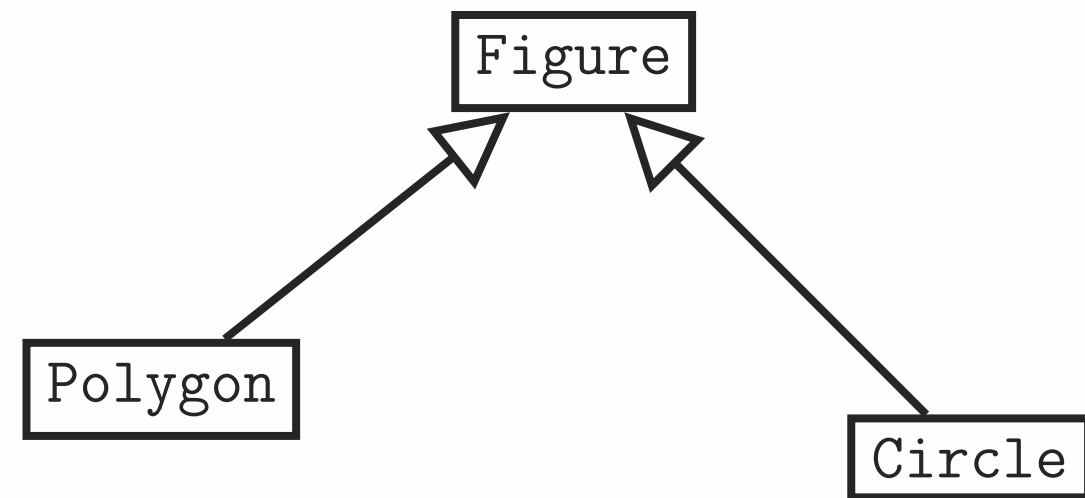
“Functions that use pointers or references to base classes must be able to use objects of derived classes without knowing it.”

— Robert C. Martin. *Clean Code: A Handbook of Agile Software Craftsmanship*. Pearson Education, 2008. DOI: [10.5555/1388398](https://doi.org/10.5555/1388398)

[ LSP SOLID Subtyping Generics Overloading ]

## Subtyping

```
1 interface Figure
2     float area();
3
4 interface Circle extends Figure
5     float perimeter();
6
7 interface Polygon extends Figure
8     int sides();
9
10 void paint(Figure f)
11     float s = f.area();
12     // ...
```



Circle  $\sqsubseteq$  Figure

Circle <: Figure

[ LSP SOLID Subtyping [Generics](#) Overloading ]

## Parametric Polymorphism (Generics)

```
1 class StackOfStrings {  
2     void push(String str) // ...  
3     String pop() // ...  
4  
5 class StackOfIntegers {  
6     void push(Integer num) // ...  
7     Integer pop() // ...  
8  
9 var s1 = new StackOfStrings();  
10 s1.push("Hello, world!");  
11  
12 var s2 = new StackOfIntegers();  
13 s2.push(42);
```

```
1 class <T> Stack<T> {  
2     void push(T item) // ...  
3     T pop() // ...  
4 }  
5  
6 var s1 = new Stack<String>();  
7 s1.push("Hello, world!");  
8  
9 var s2 = new Stack<Integer>();  
10 s2.push(42);
```

[ LSP SOLID Subtyping Generics [Overloading](#) ]

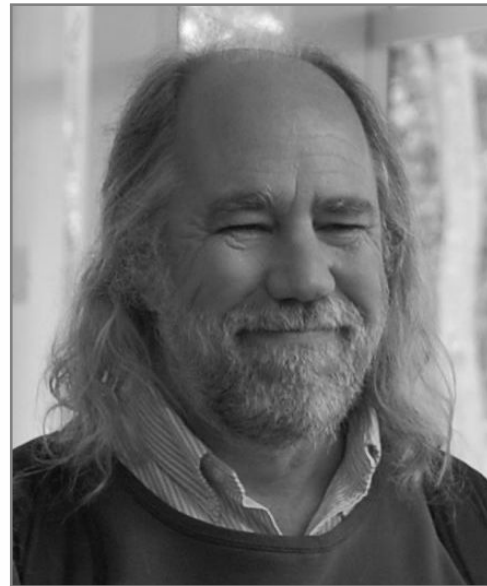
## Ad Hoc Polymorphism (Method Overloading)

```
1 class Cart {  
2     void add(int pid) // ...  
3     void addString(String pid) {  
4         this.add(Integer.parseInt(pid))  
5     }  
6 }  
7  
8 var c = new Cart();  
9 c.add(42);  
10 c.addString("17");  
11 c.addString("Hello, world!");
```

```
1 class Cart {  
2     void add(int pid) // ...  
3     void add(String pid) {  
4         this.add(Integer.parseInt(pid))  
5     }  
6 }  
7  
8 var c = new Cart();  
9 c.add(42);  
10 c.add("17");  
11 c.add("Hello, world!");
```

Chapter #2:

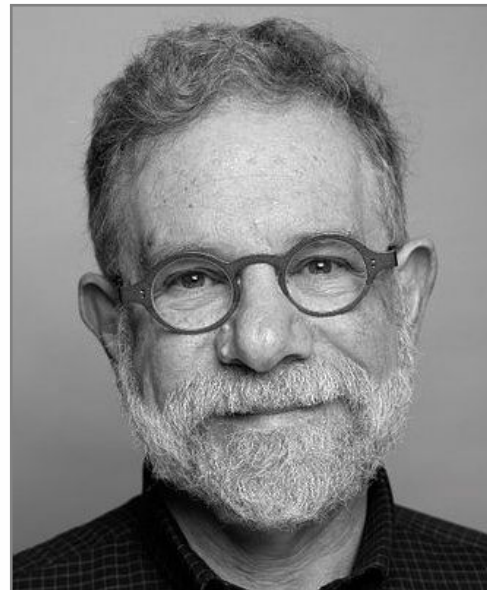
# Implementation Inheritance



GRADY BOOCH

“However, there is tension between the concepts of coupling and inheritance because inheritance introduces significant coupling. On the one hand, weakly coupled classes are desirable; on the other hand, inheritance—which tightly couples superclasses and their subclasses—helps us to exploit the commonality among abstractions.”

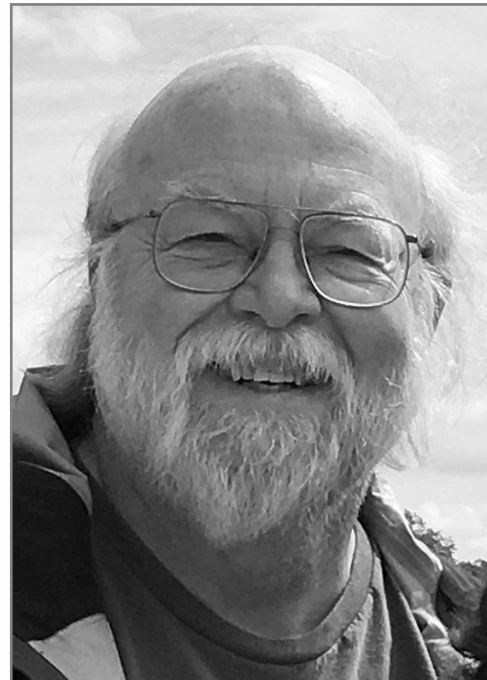
— Grady Booch et al. *Object-Oriented Analysis and Design With Applications*. Addison-Wesley, 1994. DOI: [10.5555/1407387](https://doi.org/10.5555/1407387)



“The `extends` keyword is evil; maybe not at the Charles Manson level, but bad enough that it should be shunned whenever possible.”

— Allen Holub. *Why Extends Is Evil*.

<https://jttu.net/holub2003extends>. [Online; accessed 12-09-2024]. Sept. 2003



“Someone asked him: “If you could do Java over again, what would you change?” “I’d leave out classes,” he replied.”

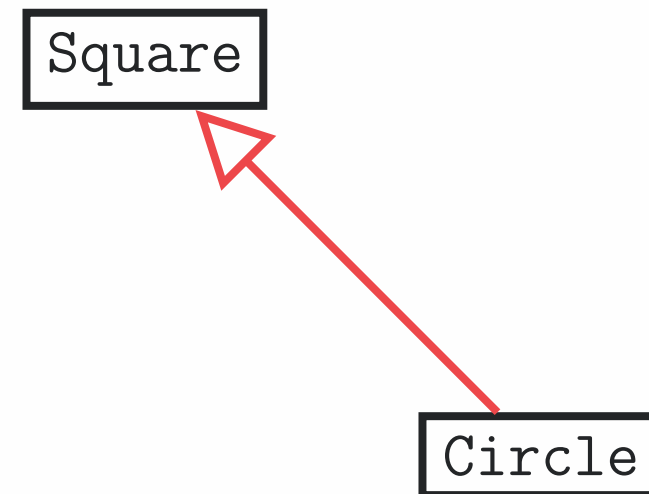
— Allen Holub. *Why Extends Is Evil*.

<https://jttu.net/holub2003extends>. [Online; accessed 12-09-2024]. Sept. 2003

[ Reuse Composition Multiple Parents ]

## Code Reuse

```
1 class Square
2     private float width;
3     float area()
4         return width * width;
5
6 class Circle extends Square
7     Circle(float radius)
8         super(radius);
9     @Override float area()
10         return 3.14 * super.area();
```



Here, the `Circle` is not a `Square`. It merely reuses the code that was negligently left open in the `Square`.

[ Reuse Composition Multiple Parents ]

Inheriting means “receive (money, property, or a title) as an heir at the death of the previous holder.” Who is dead, you ask? An object is dead if it allows other objects to inherit its encapsulated code and data.

[ Reuse Composition Multiple Parents ]

## Composition over Inheritance

### Implementation Inheritance:

```
1 class Square
2     private float width;
3     float area()
4         return width * width;
5
6 class Circle extends Square
7     Circle(float radius)
8         super(radius);
9     @Override float area()
10        return 3.14 * super.area();
```

### Composition:

```
1 final class Square
2     private float width;
3     float area()
4         return width * width;
5
6 final class Circle
7     private Square s;
8     Circle(float radius)
9         this.s = new Square(radius);
10    float area()
11        return 3.14 * s.area();
```

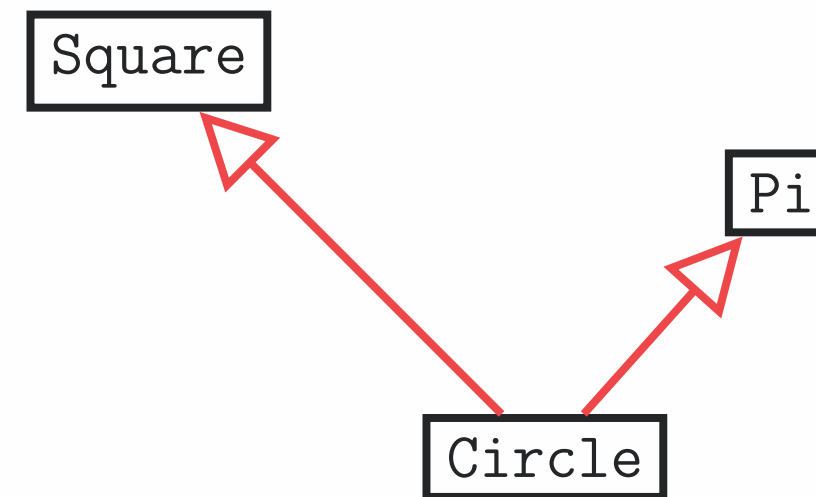
[ Reuse Composition Multiple Parents ]

All classes, without exceptions, should be either `final` or `abstract`

[ Reuse Composition Multiple Parents ]

## Multiple Inheritance

```
1 class Pi
2     float value()
3         return 3.1415926;
4
5 class Square
6     private float width;
7     float area()
8         return width * width;
9
10 class Circle extends Square, Pi
11     Circle(float r): Square(r), Pi()
12     {}
13     virtual float area()
14         return Pi.value() * Square.area
15         ();
```

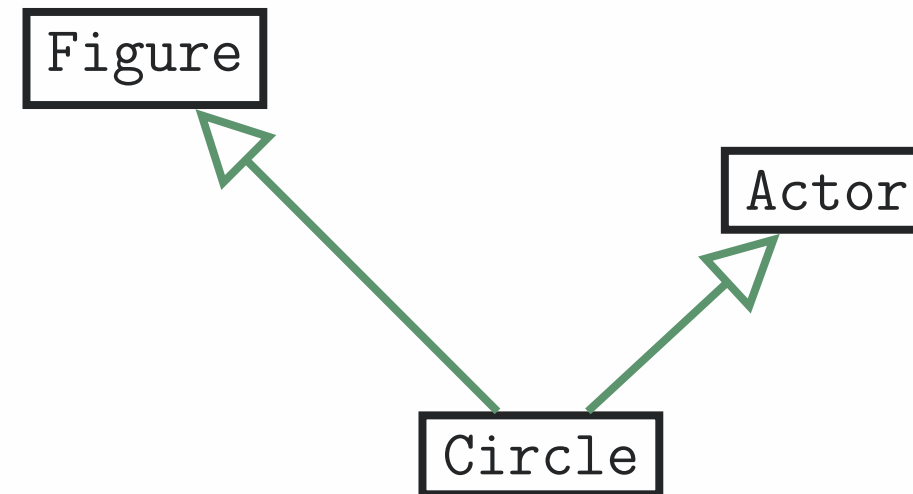


[ Reuse Composition Multiple Parents ]

[ Reuse Composition Multiple Parents ]

## Multiple Super Types

```
1 interface Actor
2     void move(int dx, int dy);
3
4 interface Figure
5     float area();
6
7 class Circle implements Figure,
8     Actor
9     Circle(float r)
10    @Override float area()
11        // ...
12    @Override void move(int dx, int
13        dy)
14        // ...
```



[ Reuse Composition Multiple Parents ]

[ Reuse Composition Multiple Parents ]

## References

- [1] Grady Booch et al. *Object-Oriented Analysis and Design With Applications*. Addison-Wesley, 1994. DOI: [10.5555/1407387](https://doi.org/10.5555/1407387).
- [2] Allen Holub. *Why Extends Is Evil*. <https://jttu.net/holub2003extends>. [Online; accessed 12-09-2024]. Sept. 2003.
- [3] Barbara Liskov. *Keynote Address — Data Abstraction and Hierarchy*. Proceedings of the Object-Oriented Programming Systems, Languages and Applications (Addendum), 1987.
- [4] Robert C. Martin. *Clean Code: A Handbook of Agile Software Craftsmanship*. Pearson Education, 2008. DOI: [10.5555/1388398](https://doi.org/10.5555/1388398).